Derivative Formula Sheet

1. $\frac{d}{dx}x^n = nx^{n-1}$
2. $\frac{d}{dx}\left(fg\right) = fg' + gf'$
3. $\frac{d}{dx}\left(\frac{f}{g}\right) = \frac{gf' - fg'}{g^2}$
4. $\frac{d}{dx}\left(f\left(g\left(x\right)\right)\right) = f'\left(g\left(x\right)\right)g'\left(x\right)$
5. $\frac{d}{dx}\left(\sin x\right) = \cos x$
6. $\frac{d}{dx}\left(\cos x\right) = -\sin x$
7. $\frac{d}{dx}\left(\tan x\right) = \sec^2 x$
8. $\frac{d}{dx}\left(\cot x\right) = -\csc^2 x$
9. $\frac{d}{dx}\left(\sec x\right) = \sec x \tan x$
10. $\frac{d}{dx}\left(\csc x\right) = -\csc x \cot x$
11. $\frac{d}{dx}\left(e^x\right) = e^x$
12. $\frac{d}{dx}\left(a^x\right) = a^x \ln a$
13. $\frac{d}{dx}\left(\ln x\right) = \frac{1}{x}$